

# OMIS 105

## Welcome Notebooks & Tech Stack

Quarter: Fall 2026

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# Tech Stack

- **Language:** Python · SQL
- **Database:** DuckDB (in-memory)
- **Notebooks:** Marimo (reactive)
- **Audience:** Senior business students
- Zero prior exposure to notebooks, SQL, or databases

# Purpose

Day-one onboarding notebooks for

## **OMIS 105 — Introduction to Database Management Systems**

These are among the first things students open.

- Introduces Marimo
- Introduces the concept of a database
- Walks through the very first SQL queries

👉 Two self-guided, interactive notebooks

# Where to Find the Notebooks

`weekly_lectures/week01-database-foundations/sql_notebooks/` :

| File                                   | Purpose                                                      |
|----------------------------------------|--------------------------------------------------------------|
| <code>SQL_Notebook_01_marimo.py</code> | Day-one student notebook — first table, first queries        |
| <code>SQL_Notebook_02_marimo.py</code> | Day-one student notebook — broader SQL 101 tour, with charts |

# Notebook 1: Structure (1/2)

1. **What is a notebook?** — Cells, text vs SQL, Cmd/Ctrl+Enter
2. **What is a database?** — Tables = spreadsheets with rows/columns
3. **First table:** `students` — CREATE TABLE + INSERT, 7 rows
4. **Asking questions with SQL** — three business questions:
  - "Who are the Marketing majors?" → WHERE
  - "Who likes Pizza?" → WHERE (different column)
  - "How many students in each major?" → GROUP BY + COUNT

## Notebook 1: Structure (2/2)

- 5. **Try It Yourself** — editable SQL cell, guided suggestions
- 6. **Why Marimo is reactive** — automatic cell updates explained
- 7. **Course roadmap** — 10-week overview table

# Notebook 2: SQL 101 Tour

A broader, denser walkthrough — better for a second sitting or self-paced review than a first-day walkthrough:

- Creating tables & inserting data
- Basic SELECT, filtering with WHERE/IN
- Sorting & LIMIT, aggregate functions, GROUP BY
- Data modification: INSERT, UPDATE, DELETE
- Charts via `plot_util.py` ( `display_result` , `plot_bar` , `plot_hbar` , `plot_pie` , `plot_line` )

# Marimo Conventions (`con.execute()`)

- The connection is created with `duckdb.connect(database=':memory:')`, and that cell **returns** `(con,)`
- Query cells: `con.execute("""...""").fetchdf()` displays the result
- Every cell touching the database takes `con` as a parameter (`def _(con):`) — that's what wires it into Marimo's reactivity
- Markdown cells use `mo.md("""...""")` with `hide_code=True`
- Use `--` SQL comments inside SQL string literals, not Python `#`
- `CREATE OR REPLACE TABLE` for re-runnability



# Notebook 1: Design Decisions (1/2)

- **Favorite foods, not business data.**

Low-stakes data (Pizza, Sushi, Tacos) keeps focus on the tool, not the business scenario. Business data starts right after, in the Week 1 demos.

- **Only a handful of rows.** Small enough to see everything at a glance.
- **Three queries only.** Enough to show the pattern —  
SELECT + FROM + WHERE, then GROUP BY.

# Notebook 1: Design Decisions (2/2)

- **"Try It Yourself" cell.**  
Hands-on editing builds comfort with the tool.  
Specific suggestions lower the barrier.
- **No plots.** Notebook 2 is where charts appear.  
Day one is about reading tables and writing SQL.

# Teaching Notes

- **In-class usage:** Open Notebook 1 live. Walk through the first few cells together (5 min), then give students 10 minutes to edit "Try It Yourself" on their own laptops.
- **Reactivity demo:** After editing, scroll back up to show other cells didn't break. "Marimo keeps everything consistent."
- **Common question:** "Where is the data stored?"  
→ In memory only. Data disappears when the notebook closes.
- **Notebook 2** is denser — better suited to a second sitting or self-paced review than the first-day walkthrough.

# Related Materials

| Folder                                                    | Content                                                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <code>weekly_lectures/week01-database-foundations/</code> | The welcome notebooks above, plus demo1–demo6                                             |
| <code>weekly_reviews/</code>                              | One self-contained review notebook per week ( <code>week01</code> – <code>week10</code> ) |
| <code>software_installation/</code>                       | Install guides, setup script, verification                                                |
| <code>tutorials/</code>                                   | Standalone SQL, DuckDB, and Git walkthroughs                                              |
| <code>data_stories/</code>                                | Standalone Python + DuckDB demo notebooks                                                 |

# Let's Get Started

*OMIS 105 — Introduction to Database Management Systems — Fall 2026*